UNIVERSITY OF PERADENIYA
Faculty of Engineering

END OF SEMESTER EXAMINATION – FEBRUARY 2019

EE 354 POWER ENGINEERING
(Two hours)

(Answer ALL Questions.)

Question 1 [40 Marks]

Bus voltages (in p.u.) of a generator after a fault are given as $V_R = 0.1\angle 0°$, $V_B = 1.1\angle 150°$ and $V_Y = 1.1\angle -150°$.

1.1 Draw phasor diagram for voltages before and after the fault. *(05 marks)*

1.2 With usual notation, write down the elements of **A** matrix. *(05 marks)*

1.3 Given that the positive, negative and the zero sequence impedances of the generator are 0.1j, 0.1j and 0.05j respectively, calculate sequence voltage and sequence currents. *(10 marks)*

1.4 Given that the base voltage and the base power are 132 kV and 100 MVA respectively, calculate actual line currents of the generator after the fault. *(10 marks)*

1.5 With justification, state the type of the fault and find the fault current in Amperes. *(10 marks)*

Question 2 [60 marks]

Following information are given for a five bus system for a fault analysis:

Generator 1	Connected to Bus 1; Star connected and solidly grounded; Positive, negative and zero sequence impedances are 0.1j, 0.1j and 0.05j respectively.
Generator 2	Connected to Bus 4; Star connected and solidly grounded; Positive, negative and zero sequence impedances are 0.1j, 0.1j and 0.05j respectively.
Transformer 1	Two winding; Primary side is star connected, solidly grounded and connected to bus 1; Secondary side is delta connected and connected to bus 2, transformer impedance is 0.1j.
Transformer 2	Three winding; Primary side is star connected, solidly grounded and connected to bus 3. Impedance is 0.05j; Secondary side is star connected, solidly grounded, and connected to bus 4. Impedance is 0.05j; Tertiary side is star connected, solidly grounded, and connected to bus 5. Impedance is 0.05j.
Transmission line	Connected to buses 2 and 3; Positive, negative and zero sequence impedances are 0.1j, 0.1j and 0.2j respectively

2.1 Draw the circuit diagram for the bus system and mark all information. *(05 marks)*

2.2 Draw the positive, negative and the zero sequence networks for this bus system. *(15 marks)*

2.3 If a fault occurs in bus 5, draw the simplified sequence networks to represent the single generator model. *(15 marks)*

2.4 Calculate the positive sequence current, if the fault is:

a) Single line to ground in R phase

b) Double line fault between Y and B phases

c) Double line to ground fault in Y and B phases

d) Three phase fault

(25 marks)

Question 3 [50 Marks]

3.1 A synchronous motor is to be used for power factor correction.

a) Draw the v-curve of a synchronous motor for two loads (P_1 and P_2 where $P_1 > P_2$). Indicate the important regions.

b) What is a synchronous condenser?

c) Sketch the phasor diagram of a synchronous condenser.

(16 marks)

3.2 A 230 V, 50 kVA, 0.85 lagging, Δ-connected, 50 Hz synchronous motor is connected to a 32 hp load. The synchronous reactance is 2.75 Ω and the armature resistance is negligible. The rotational losses of the motor are 3.5 kW.

a) Calculate the internally generated voltage. Assume that the motor is operating at 0.82 power factor lagging.

b) Calculate the phase current if the load on the motor is increased to 40 hp.

(20 marks)

3.3 Explain the reduced frequency starting method used for synchronous motors. State the safety precautions. Further explain how this method can be practically implemented. *(14 marks)*

Question 4 [50 Marks]

4.1 Consider a salient-pole synchronous generator

a) Sketch the phasor diagram.

b) Explain with illustrations how to construct the phasor diagram when the stator supply voltage and current measurements are available.

(16 marks)

4.2 A synchronous generator is short circuited due to a fault.

a) State the two components that can be identified in the fault current in a phase.

b) Write an equation for the rms magnitude of the ac fault current with usual notations.

c) Specifications of the generator are 100 MVA, 13.5 kV, Y-connected, three-phase and 60 Hz. The generator is operating at the rated voltage and no-load before a three-phase fault occurs at its terminals. The reactances of the machine in per unit (to machines own base) are $X_s = 1.0$, $X' = 0.30$ and $X'' = 0.15$ respectively. The time constants are $T' = 1.15\ s$ and $T'' = 0.03\ s$ respectively. Reactances and time constants are indicated with usual notations. Obtain the equation for the rms magnitude of the ac faculty current.

(24 marks)

4.3 Discuss the fixed-speed wind turbine variants along with their advantages and disadvantages.

(10 marks)

ENGINEERING LIBRARY
12 MAR 2020
UNIVERSITY OF PERADENIYA